

The IL-23R–p19 interface as a small-molecule drug target in psoriasis: structural basis, discovery strategies, and preclinical target product profile

Target assessment report

September 7, 2026

Abstract

Psoriasis affects approximately 125 million people worldwide, and the interleukin-23 (IL-23)/IL-23 receptor (IL-23R) axis is one of its most validated drug targets: three subcutaneous anti-IL-23 monoclonal antibodies (mAbs) achieve complete or near-complete skin clearance (investigator’s global assessment [IGA] 0/1 or static physician’s global assessment [sPGA] 0/1) in 80–85% of patients by week 16. Yet no small-molecule inhibitor of this axis is in the clinic. This report assesses the IL-23R–p19 interface, where the p19 (IL-23A) subunit of IL-23 binds the immunoglobulin domain 1 (D1) of IL-23R, as a small-molecule target. Three findings drive the assessment. First, the structural white space is absolute: an exhaustive survey of the Protein Data Bank (PDB) and PubMed found, in the retrievable literature, no structure of a small molecule bound to IL-23 or IL-23R; the p19–IL-23R D1 interface is a flat, pocketless surface of about 900 Å² that a small molecule must occlude. Second, protein tools have mapped the interface completely—nanobodies, a single-domain antibody (VHH) with a dissociation constant (K_D) of 128 pM, and cyclic peptides as potent as 28 nM—and one of them, icotrokinra (an oral macrocyclic peptide; phase 3 results reported; approval status unverified), has shown that direct oral antagonism of IL-23R is clinically achievable, reaching 68–70% IGA 0/1 by week 16. No oral small molecule has achieved this. Third, the gap is addressable by structure-based design: five ranked strategies target the W156 hotspot cluster, the p19-only contact geometry, and the 2025-reported induced pocket on p19. A preclinical target product profile (TPP) sets the bar: week-16 sPGA 0/1 $\geq 80\%$ (the mAb bar), onset by week 4, oral once-daily dosing with a 12–24 h half-life, serious infections $\leq \sim 1.5$ per 100 patient-years, no impact on IL-12, and ≥ 10 –30-fold selectivity over interferon- γ (IFN γ). The risks are real—human–mouse cross-reactivity is unverified and no in-vivo small-molecule data exist—but the total absence of a small-molecule structure at this interface is a first-mover opportunity, not evidence of inaccessibility.

1 Introduction

1.1 Disease burden and genetics

Psoriasis is a chronic immune-mediated skin disease affecting approximately 125 million people worldwide [1]. In the United States, the NHANES 2011–2014 survey found a 3.0% prevalence

among adults, corresponding to roughly 7.55 million patients [2]. Plaque psoriasis is the most common form. The disease carries substantial comorbidity: psoriatic arthritis co-occurs in 19.7% of patients, depressive symptoms are present in 28%, and cardiovascular risk carries an odds ratio of 1.27 relative to the general population [1].

Genetically, psoriasis is one of the most polygenic complex diseases: genome-wide association studies implicate 63 loci that explain about 28% of heritability (h^2) [47]. HLA-C*06:02 is the principal risk allele and a biomarker of biologic response, and the PSORS1 locus encodes caspase recruitment domain-containing serine protease (CDSN). For target validation specifically, the protective IL-23R variant R381Q (rs11209026), reported in genome-wide association studies as reducing the risk of both psoriasis and inflammatory bowel disease (IBD) (Duerr 2006; Capon 2007), maps onto the IL-23:IL-23R complex itself [11]. A human genetic variant that attenuates IL-23R signaling is protective: this is direct evidence that blocking this interface is both an effective and a tolerable disease intervention.

1.2 The Th17 checkpoint

IL-23 is a heterodimer of p19 (encoded by *IL23A*) and p40 (encoded by *IL12B*, shared with IL-12). It signals through a receptor composed of the IL-23-specific subunit IL-23R and the shared subunit IL-12R β 1 (shared with IL-12). Engagement is two-tiered: p19 binds IL-23R D1 only, and this event recruits IL-12R β 1 with roughly 100-fold cooperative strengthening (binary affinity $\approx 2 \mu\text{M}$; recruitment $\approx 25 \text{ nM}$; the IL-23:IL-23R affinity is $\sim 10\text{--}44 \text{ nM}$, method-dependent [11, 27]). Receptor engagement activates the kinases JAK1/JAK2/TYK2 and the transcription factors STAT3 and STAT4, sustaining the pathogenic T-helper 17 (Th17) cell state and the IL-17A/F, IL-22, GM-CSF, and CCL20 cytokine network that drives keratinocyte expansion and inflammation [22].

Two further biology findings matter for a drug program. First, the surface receptor is shed: phorbol-12-myristate-13-acetate (PMA)-stimulated ADAM17, and constitutive ADAM10, cleave IL-23R, generating soluble IL-23R (sIL-23R) whose serum levels track disease activity—a candidate pharmacodynamic (PD) and monitoring biomarker [18]. Second, the depth of suppression required for clinical response is high: single-cell analysis in psoriasis patients shows that responders require $>90\%$ suppression of IL-17 signaling in lymphocyte populations adjacent to keratinocytes [52]. The target is therefore validated by genetics, pathway position, shed-receptor biomarker, and depth-of-response data. The open question is chemical: how to block a flat protein–protein interaction (PPI) with a small molecule, orally, at mAb-class safety.

2 Structural basis of the IL-23/IL-23R complex

2.1 A two-tier, p19-centric interface

The quaternary signaling complex (IL-23 + IL-23R + IL-12R β 1) is solved at 3.4 Å resolution in PDB entry 6WDQ [19] (Figure 1). Its topology is non-canonical: the two receptor subunits do not contact each other (IL-23R and IL-12R β 1 never approach within 11.5 Å), and p19 never contacts IL-12R β 1 (minimum distance 23.5 Å; zero atom pairs within contact distance in 6WDQ). The entire IL-23:receptor contact surface is split between two sites. p40 simultaneously engages IL-23R D1 (about 260 Å²) and IL-12R β 1 (about 1532 Å²) with strong charge complementarity—p40 FG-loop His216/Lys217/Lys219 and Trp37/Phe82 facing β 1 Glu28/Asp58/Asp101 plus a polar ring (Gln102/Ser106/Tyr109/Gln132/Tyr134)—while p19 engages only IL-23R D1 (about 900 Å², 51 residues, mixed hydrophobic and charged character).

For drug discovery this is the defining architectural feature: the p19:IL-23R D1 contact is a discrete, subunit-specific site that a small molecule can occupy without touching the IL-12:IL-12R β 1

machinery. A p19-selective ligand does not need to occupy the shared p40:IL-12R β 1 arm at all.

2.2 Structural catalog

Table 1 lists the IL-23/IL-23R structures available for design. The lead structure for interface design is [5MZV](#), the first IL-23:IL-23R complex; [6WDQ](#) defines the quaternary signaling geometry; [8CR8](#) is the highest-resolution apo IL-23; and the two 2025 structures, [8UUI](#) and [9MFG](#), both demonstrate that new binding sites can be built on this system—an induced pocket on p19 and a VHH on IL-23R D1, respectively.

Table 1. PDB catalog for IL-23/IL-23R structure-based design.

PDB	Resolution	Content	Reference
6WDQ	3.4 Å	Quaternary IL-23/IL-23R/IL-12R β 1 signaling complex; p19 CDR3 disordered; two glycans	Glassman 2021 [19]
5MZV	2.8 Å	First IL-23:IL-23R structure (with nanobody Nb22E11); binary complex geometry	Bloch 2018 [11]
8CR8	2.0 Å	Apo human IL-23, highest resolution	Bloch 2024 [12]
8OE4	3.6 Å (cryo-EM)	Pre-dimerized, complete extracellular signaling complex	Bloch 2024 [12]
8UUI	2.43 Å	p19+p40 with peptide 23-446 in an induced pocket	Lecomte 2025 [27]
9MFG	2.85 Å	IL-23R with protease-stable VHH (oral single-domain antibody)	Ota 2025 [31]
4GRW	2.55 Å	Dimeric IL-23 with three nanobody dimers engaging p19 and p40	Desmyter 2017 [14]
6UIB	2.74 Å	IL-23 with cyclic peptide 23-652 (first cyclic peptide on IL-23p19)	Pandya 2020 [32]
3DUH	2.3 Å	Human IL-23 (first IL-23 structures, 2008)	Lupardus 2008 [29]
3D87	2.9 Å	Human IL-23 (first IL-23 structures, 2008)	Beyer 2008 [5]

Context structures not tabulated: [3D85](#) (1.9 Å, IL-23 with Fab 7G10; [[5](#)]), the [3QWR](#) adnectin, the [5MJ3](#)/[5MJ4](#) alphabody [[13](#)], [5NJD](#) (briakinumab; [[11](#)]), and [5MXA](#) (apo IL-23R D1; [[11](#)]). Verified negatives, from an exhaustive PDB and PubMed sweep: no small-molecule + IL-23 or IL-23R complex exists in the PDB (all IL-23 entries carry empty ligand lists); no standalone human IL-23R ectodomain is deposited (the closest structure is [9MFG](#), which retains the receptor in context); and no covalent inhibitor is known at this interface.

2.3 Interface geometry

A residue–residue contact map of the p19–IL-23R D1 interface computed from [5MZV](#) (272 atom pairs within 5 Å) resolves into two cluster blocks: p19 residues 53–60 (CDR1) and 154–165 engaging IL-23R residues 24–31 (primary cluster) and 98–118 (secondary cluster) ([Figure 2](#)). In the quaternary complex [6WDQ](#) the contact census is p19–IL-23R 1234 pairs versus p40–IL-23R 223; p40–p19 1249; p40–IL-12R β 1 850; and p19–IL-12R β 1 zero—the receptor subunits do not directly contact, and the IL-12R β 1 arm is recruited through p40 only.

2.4 Hotspot residues

At residue level, the [5MZV](#) interface is dominated by a small set of high-impact contacts ([Table 2](#) and [Fig. 3](#)). Trp156 (W156), at the N-terminus of p19 helix D, forms a hydrogen bond to the Asp118 (D118) carboxylate (2.3 Å; D118 donating to the W156 indole N) and stacks against Gly116 (G116)

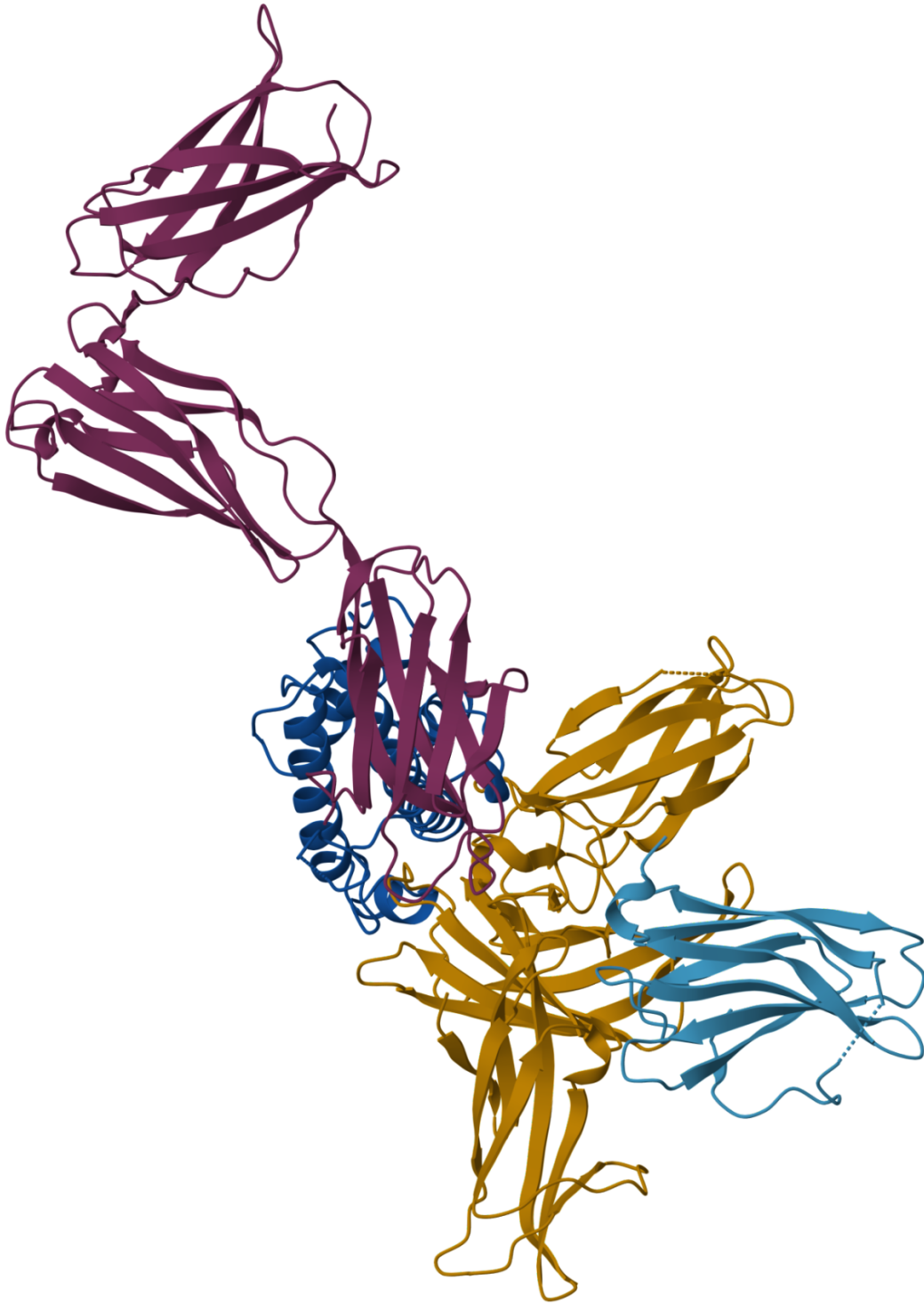


Figure 1. Quaternary signaling complex IL-23/IL-23R/IL-12R β 1 (PDB [6WDQ](#), 3.4 Å; Glassman et al., *Cell* 2021 [19]): p40 (gold) anchors IL-23R D1 and IL-12R β 1 (cyan); p19 (blue) engages only IL-23R (magenta) D1.

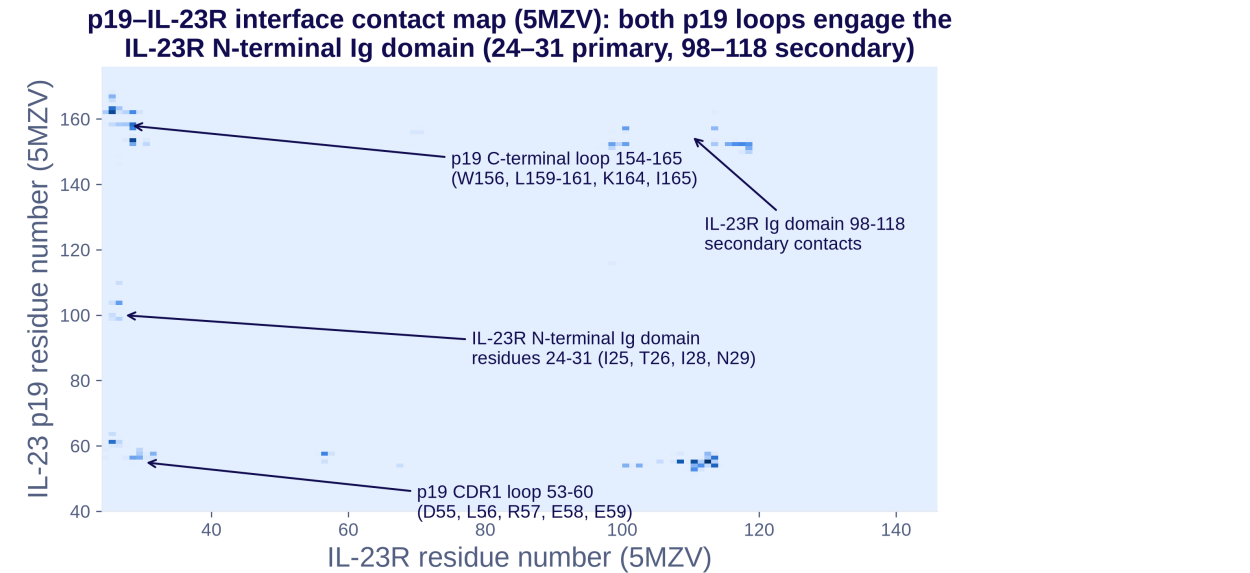


Figure 2. Residue–residue contact map of the p19–IL-23R D1 interface computed from PDB 5MZV (2.8 Å; Bloch et al., *Immunity* 2018 [11]). The 272 atom pairs within 5 Å define two cluster blocks: p19 53–60 (CDR1) and 154–165 engaging IL-23R 24–31 (primary) and 98–118 (secondary).

at 2.7 Å; G116 sits in the Ile-Cys-Gly cassette shared with IL-12R β 2. Lys164 (K164) of the CDR1 loop makes a 1.7 Å contact with the D1 N-hook Gly24 (G24) and hydrogen-bonds the G24/N27 main-chain carbonyls, a network stabilized by p19 Glu58 (E58). The D1 N-hook (G24–N29, five residues of a non-canonical Ig domain tethered by the C30–C115 disulfide) is what orders the p19 AB loop. The Cys73(p19)–Cys199(p40) disulfide (2.0 Å) links the two cytokine subunits; it is a structural fact, not a drug target.

Mouse alanine scans (surface plasmon resonance/BLI) converge on the same hotspots: murine W157A/L161E (approximating human W156/L160) abolish binding, whereas K165S/L56E retain it [11].

Table 2. Residue-level interface between p19 and IL-23R D1, from PDB 5MZV (2.8 Å; Bloch et al. 2018 [11]).

p19 residue	IL-23R D1 residue	Interaction	Distance	Role
W156 (helix D)	D118	H-bond (W156 indole N)	2.3 Å	Primary hotspot
W156 (helix D)	G116	Stacking	2.7 Å	Primary hotspot
K164 (CDR1)	G24 (N-hook)	H-bond network (K164 → G24/N27 main-chain C=O)	1.7 Å	CDR1 anchor
E58 (CDR1)	(D1)	Stabilizes K164–G24 network	—	Supporting
Cys73	Cys199 (p40)	Disulfide	2.0 Å	Structural only (cytokine link)

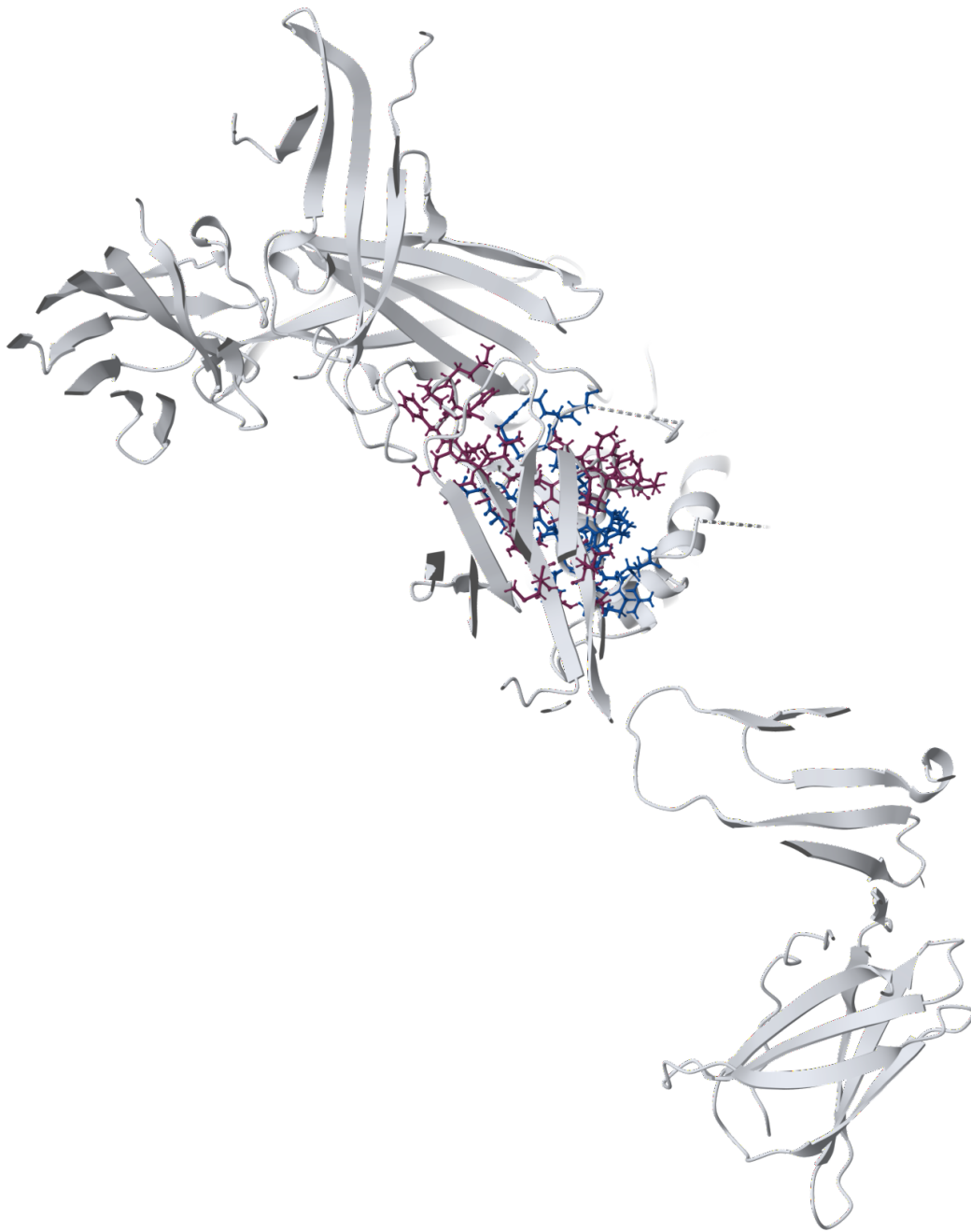


Figure 3. Close-up of the p19-IL-23R D1 interface (PDB [5MZV](#), 2.8 Å; Bloch et al., *Immunity* 2018 [11]): contact clusters p19 53–60 (CDR1) and 154–165 (blue) against IL-23R 24–31 (N-hook) and 98–118 (magenta), with the W156 hotspot region at the center of the interface.

2.5 Flexibility and induced fit

The interface is plastic. IL-23R rotates about 18° between the binary complex [5MZV](#) and the quaternary complex [6WDQ](#), and binding of IL-23R converts the N-terminus of p19 helix D from an α -helix to a 3_{10} helix, projecting W156 toward D1 [[11](#)]. Apo IL-23 ([5MXA](#), [8CR8](#)) is more flexible, and receptor binding primes and locks the interface. Apo structure [8CR8](#) is shown in [Figure 4](#).

The 2025 structure [8UUI](#) adds the decisive small-molecule-relevant observation: a cryptic pocket is inducible on p19. Peptide 23-446 (a 20-residue peptide, [[27](#)]) displaces residues 138–153 of the p19 CD loop/CDR3 region (disordered in that structure) and induces the pocket it occupies. Pockets need not be preformed; the p19 surface can be deformed by its binder, and the disordered CDR3 CD loop is the softest region.

2.6 Caveats

Two caveats condition every design decision downstream. First, glycosylation: IL-23R carries seven potential N-glycosylation sites (N29, N47, N81 in D1; N141, N180, N232, N262 in D2/D3), and no N-acetylglucosamine is modeled on IL-23R in any deposited complex ([5MZV](#), [6WDQ](#), [8OE4](#)). The role of the N29 glycan at the D1:p19 interface is uncertain (UNCERTAIN) and is the key caveat for receptor-side small molecules. Second, the CDR3 boundary (approximately residues 135–154) is a derived, not a structure-defined, range; the disordered region observed in [8UUI](#) (residues 138–153) is the structure-verified portion.

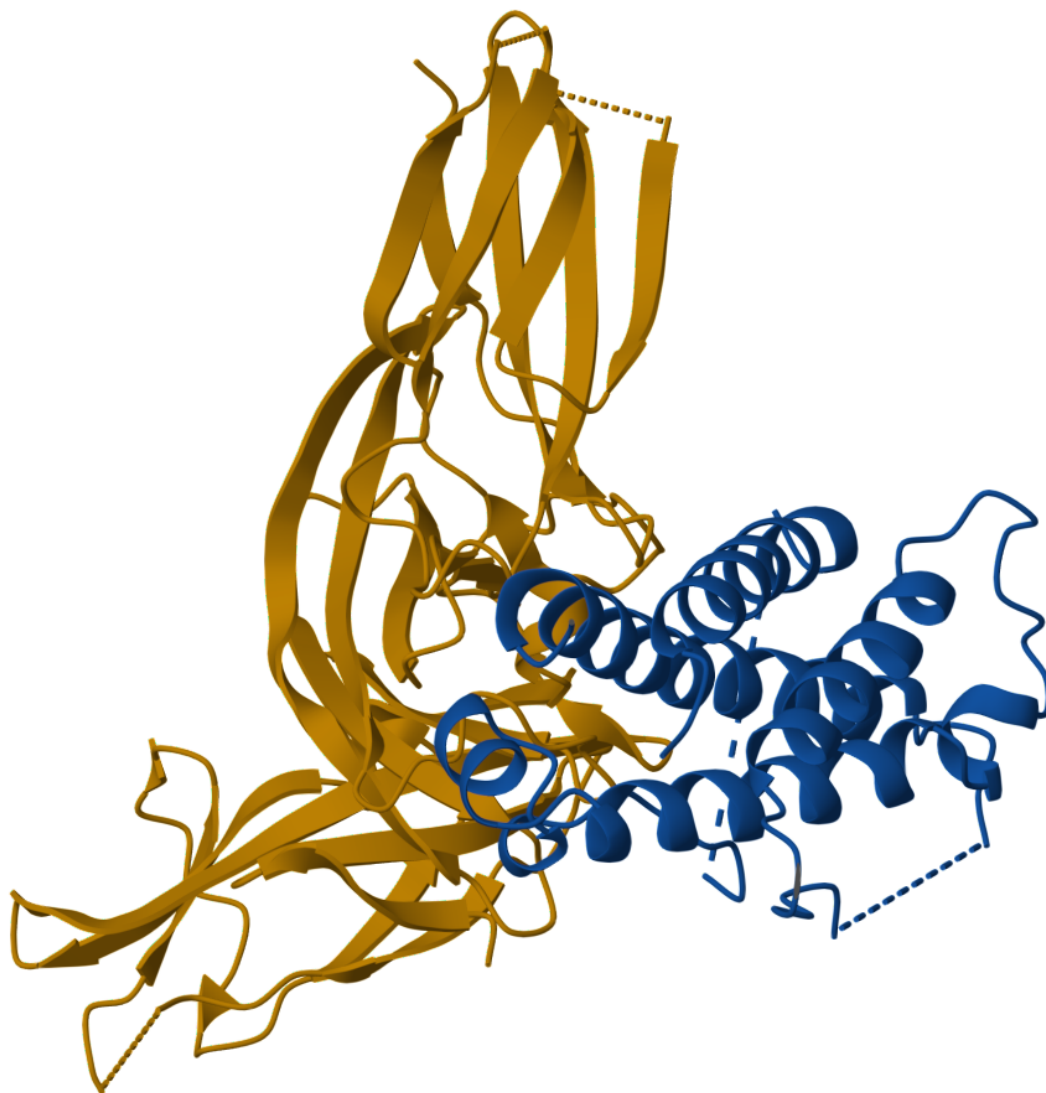


Figure 4. Apo human IL-23 (PDB [8CR8](#), 2.0 Å; Bloch et al., *Nature Structural & Molecular Biology* 2024 [[12](#)]): p40 (gold) + p19 (blue, four-helix bundle). The unliganded p19 surface shows no preformed pocket at the future IL-23R contact site.

3 The hard-PPI problem and the protein tooling landscape

3.1 Why small molecules struggle here

Five properties make this one of the harder PPI targets in cytokine biology:

1. **Flatness.** About 900 Å² of p19 contact over 51 residues, with no preformed pocket at the IL-23R contact site in the apo structure (Figure 4).
2. **Affinity to beat.** K_D of 10–44 nM for IL-23:IL-23R (5.2–10.8 nM by surface plasmon resonance [SPR]; ~44 nM by biolayer interferometry [BLI]; method-dependent [11, 27]). A small molecule must displace a sub-nanomolar protein ligand.
3. **Selectivity.** The p40 arm is shared with IL-12, and the Ig D1 fold is common among receptors; the only p19-specific structural information available is the p19:IL-23R D1 interface itself.
4. **Zero precedent.** No PDB structure of a small molecule at this interface exists (verified), and the best published fragments are at K_D of 180 µM–1.95 mM—proof-of-concept only [27].
5. **The comparator bar.** An oral macrocyclic peptide (icotrokinra) is in phase 3 at 68–70% week-16 IGA 0/1, with an mAb-class safety profile to match (Section 6).

The protein tools, however, have mapped the problem with unusual precision.

3.2 Nanobodies

PDB 4GRW (2.55 Å) shows dimeric IL-23 engaged simultaneously by three nanobody (Nb) dimers on epitopes spanning p19 and p40 (Figure 5, [14]); Nb22E11 in 5MZV [11] and the 5MJ3/5MJ4 alphabody [13] provide further p19 epitopes. These structures define the multivalent-engagement template and the p19 epitope boundaries that small-molecule design can imitate.

3.3 A VHH that reaches the D1:p19 site (9MFG)

The 2025 structure 9MFG (2.85 Å) captures a protease-stable VHH bound to IL-23R D1 [31]. Its K_D is ~128 pM (human) and 71 pM (cynomolgus); screening VHHs show IC₅₀ values below 30 nM in functional assays. The epitope overlaps the p19 D1 site substantially—the VHH is competitive with IL-23—with 1073 Å² of buried surface and eight side-chain hydrogen bonds (R45, S54, T56, Y59, T68, K98, W104, Q105). Functionally, it is the only oral IL-23-pathway agent in the literature: stable for >8 h in simulated intestinal fluid and >24 h in fecal fluid, and a single pill in non-human primates produces >24 h of receptor inhibition [31]. The design implication: the D1:p19 site can be occupied by an oral, protease-resistant, pM-affinity binder; the site is not off-limits, and the small-molecule version of this geometry is the open problem.

3.4 Peptides

The first cyclic peptide on IL-23p19, 23-652 (PDB 6UIB, 2.74 Å), showed low-µM affinity and inhibition against IL-23R [32]. The Ferring 2025 program [27] advanced the p19-induced pocket: peptide 23-446 (a 20-residue peptide; PDB 8UUI, 2.43 Å) binds the induced pocket (Figure 6), and structure-activity refinement produced peptide 14 (IC₅₀ 64 nM, K_D 16 nM) and peptide 15 (IC₅₀ 28 nM, K_D 20 nM)—low-nanomolar, on-target, direct-binding peptides at this interface. A parallel fragment-to-peptide pipeline: a fragment-binding high-throughput screen (HTS) against a fluorescent p19 probe yielded 353 hits (2.8% hit rate), of which 24 showed ≥50% inhibition (0.2%); the top five were confirmed by differential scanning fluorimetry and microscale thermophoresis

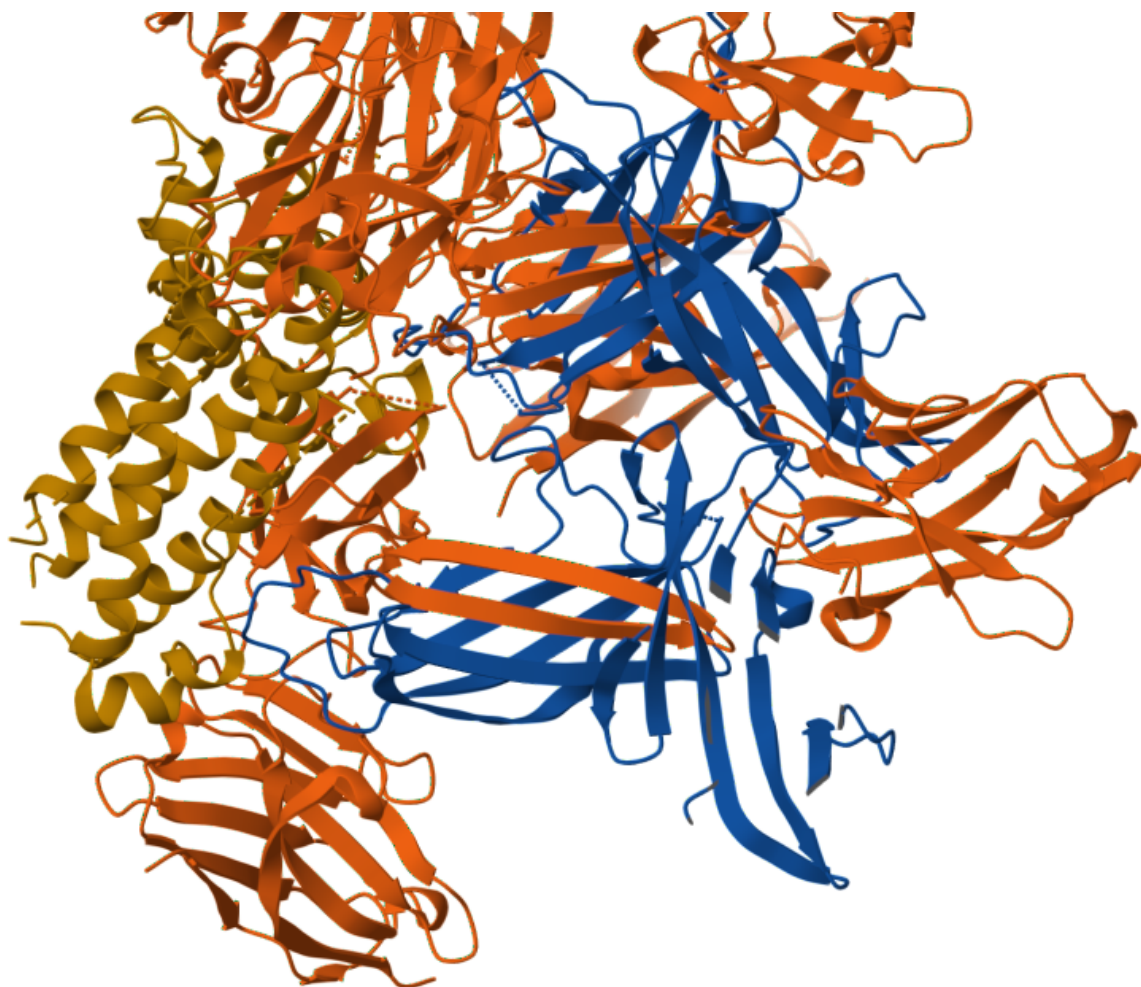


Figure 5. Dimeric IL-23 (PDB [4GRW](#), 2.55 Å; Desmyter et al., *Frontiers in Immunology* 2017 [[14](#)]) with three nanobody dimers (orange) engaging epitopes on p19 (blue) and p40 (gold) simultaneously—a template for multivalent engagement.

(DSF/MST) with K_D values of 180 μM –1.95 mM. The icotrokinra scaffold derives from de novo proteolytically resistant therapeutic peptides [24] and has been disclosed for this interface [23].



Figure 6. Induced pocket: peptide 23-446 (orange; PDB [8UUI](#), 2.43 Å; Lecomte et al., *ACS Chemical Biology* 2025 [27]) binds p19 (blue) and induces the pocket it occupies; p40 (grey) for context.

3.5 Icotrokinra: the benchmark

Icotrokinra (JNJ-77242113) is an oral macrocyclic peptide IL-23R antagonist—a peptide, not a small molecule—and is the direct comparator for any IL-23R small-molecule program. Translational data [17]: K_D 7.1 pM to IL-23R; IC_{50} 5.6 pM for proximal signaling; no measurable impact on IL-12 signaling; $\text{IFN}\gamma$ IC_{50} 9–18 pM (18.4 pM in a hidradenitis-suppurative cell line; 11 pM in psoriasis samples)—the peptide is therefore not highly selective over $\text{IFN}\gamma$, which makes selectivity a design parameter rather than an assumption. It shows oral efficacy in the rat TNBS (trinitrobenzene sulfonic acid) colitis model at ≥ 0.3 mg/kg/d and in the rat IL-23 skin model.

Clinically: FRONTIER-1 (phase 2, *NEJM* 2024 [6]; $n = 255$; 16 weeks) week-16 PASI-75 of 37% (25 mg once daily [QD]), 51% (25 mg twice daily [BID]), 58% (50 mg QD), 65% (100 mg QD), and 79% (100 mg BID) versus 9% for placebo ($P < 0.001$). FRONTIER-2 week 52 (100 mg BID [16]): PASI-75/90/100 of 76%/64%/40%; IGA 0/1 74%; IGA 0 43%. Phase 3 ICONIC-ADVANCE 1 & 2 (*Lancet* 2025 [20]; 200 mg QD versus placebo and deucravacitinib 6 mg QD): week-16 IGA 0/1 68% (213/311) versus 11%, and 70% (227/322) versus 9%; PASI-90 55%/57% versus 4%/1%; all comparisons $P < 0.0001$, superior to both comparators. Week-16 adverse events 48% versus 57% (placebo); week-24 57% versus 65% (deucravacitinib). The IBD phase 2 (NCT04102111) was terminated for futility. Phase 3 results are reported; approval status is unverified.

4 The small-molecule landscape: what has actually been done

The headline, in the words of the only published small-molecule PPI program at this interface: “no small-molecule IL23-targeting therapeutics have been identified” [27]. In the retrievable literature, four programs define the state of the art:

1. **Ferring/BTG 2025** [27]—the only published program with validated direct binding at the interface (Section 3). The fluorescent-p19 fragment HTS identified 353 hits (2.8%); the confirmed top fragments are at K_D 180 μM –1.95 mM.
2. **Janssen 2021** [49]—cell-based HTS in DT40 cells stably co-expressing IL-23R/IL-12R β 1 with a STAT5–luciferase reporter: $\sim 24,000$ compounds, 5% hit rate; 1288 primary hits, 209 after cytotoxicity and IFN γ counter-screen, and 24 with $\text{IC}_{50} < 7 \mu\text{M}$, $\geq 80\%$ inhibition, and > 3 -fold selectivity over IFN γ (tofacitinib as positive control). Direct binding to IL-23/IL-23R was not demonstrated.
3. **GSK 2023** [25]—fluorescent IL-23 (IL23-TMR) with cyclic-peptide probes in a NanoBRET assay against full-length receptor; competitive and non-competitive peptide antagonist classes identified; the disease-associated C115Y IL-23R variant disrupts the p19 epitope; a phage-selected binder, P630, binds IL-23R alone (patent US9624268B2); a macrocyclic dodecapeptide maps to the p19:IL-23R interface by hydrogen-deuterium exchange mass spectrometry (HDX-MS) [40].
4. **Thai 2026** [45]—in silico only: IL-23 + risankizumab complex reconstructed by homology/deep learning, 3D pharmacophore, DrugBank + ZINC12 docking; the top hit ZINC20572287 (“r3-7”) sits in a p19 pocket—not the IL-23R interface—and is stable over a 600-ns molecular dynamics simulation. No wet-lab validation; no potent p40-directed inhibitor found.

The white space, restated with its caveat: patent and registry databases (Google Patents, EPO Espacenet, USPTO, PatentsView, FreepatentsOnline) returned HTTP 403 at query time, so “no small-molecule IL-23R inhibitor is published” is true in the retrievable literature (PubMed, PDB, journals) and is not a complete claim over the patent world.

For any future campaign the assay hierarchy implied by the published programs is: (i) direct PPI binding (SPR/BLI/ITC [isothermal titration calorimetry]/fluorescence polarization/DSF/MST)—only Ferring has validated this tier; (ii) NanoBRET or fluorescent-IL-23 binding against full-length receptor (GSK); (iii) cell-based reporter with an IFN γ counter-screen (Janssen); and (iv) in vivo—no small molecule has published in-vivo data at this node.

5 Discovery strategies for a p19/IL-23R small molecule

Five strategies, ranked by expected value and data support:

1. **Loop-cluster targeting (primary).** The p19 CDR1 (53–60) and 154–165 loop clusters define the W156 hotspot on IL-23R D1 24–31 and 98–118 ([Figure 3](#)). Design planar-aromatic plus charged mimics that reproduce the W156/K164 pattern against D118/G116/G24/N27: aromatic stacking on G116, a hydrogen bond to the D118 carboxylate, and a K164-like interaction with the N-hook G24/N27 carbonyls.
2. **p19-only selectivity (class-first property).** p19 never contacts IL-12R β 1 (minimum 23.5 Å), so any ligand on the p19:IL-23R D1 interface is structurally blind to IL-12—unlike p40-directed or receptor-shared designs. Conformation-specific binding (the IL-23-bound p19 conformation only, or the [8UUI](#)-induced pocket only) adds a second selectivity layer.
3. **Induced-pocket exploitation.** [8UUI](#) proves a cryptic pocket is buildable on p19 by displacing residues 138–153 ([Figure 6](#)). A fragment-to-lead campaign from the published fragment series (180 μ M–1.95 mM proof-of-concept) toward the same pocket: the gap from 1.95 mM to peptide 15 (20 nM) is the headroom to be captured.
4. **Receptor-side IL-23R D1 ([9MFG](#) template).** The VHH [9MFG](#) defines the D1:p19 site at K_D 128 pM; a small molecule occupying D1 24–31/98–118 would block p19 binding from the receptor side. Caveats: N29 glycosylation is unmodeled (role uncertain; verify by glyco-engineering before committing leads), and cross-reactivity with other Ig D1 receptors must be characterized.
5. **Bivalent/multivalent anchoring with p40 assistance ([4GRW](#) template; exploratory).** Simultaneous engagement of a p19 epitope plus p40, as in the [4GRW](#) nanobody dimers, raises avidity on a flat surface; caution: p40 is shared with IL-12, so p40 engagement trades away IL-12 selectivity.

Context only, not lead strategies: the TYK2/JAK node (proven oral route, but JAK-class safety baggage—[Section 7](#)); the shed sIL-23R biomarker [[18](#)]; and IL-22, which signals through IL-22R1/IL-10R β 2 without IL-12R β 1 (no IL-22 cross-activity expected; IL-35 receptor usage uncertain—test).

6 Clinical efficacy landscape

6.1 Monoclonal antibodies (the bar)

Guselkumab (100 mg subcutaneous [SC] at weeks 0/4 then every 8 weeks; FDA/EMA 2017): VOYAGE 1 week-16 IGA 0/1 85.1% versus 6.9% placebo (adalimumab 65.9%), PASI-90 73.3% versus 2.9% [[7](#)]; VOYAGE 2 week-16 IGA 0/1 84.1% versus 8.5%, PASI-90 70.0% versus 2.4% [[37](#)].

Risankizumab (150 mg SC every 12 weeks; FDA April 2019): IMMvent week-16 PASI-90 72% versus adalimumab 47%, sPGA 0/1 84% versus 60%, and week-52 66% versus 21% [[38](#)]; IMMerge week-52 PASI-90 86.6% versus secukinumab 57.1% [[51](#)]; LIMMitless week-256 PASI-90 85.1% and PASI-100 52.3% [[34](#)]; head-to-head against deucravacitinib, IMMpactful week-16 sPGA 0/1 80.2% versus 39.7%, PASI-90 57.3% versus 22.9%, PASI-100 27.5% versus 6.5%, total adverse events 33.6% versus 42.9% [[30](#)]. Context: UltIMMa week-16 PASI-90 75.3%/74.8% versus placebo 4.9%/2.0% and ustekinumab 42.0%/47.5% [[21](#)].

Tildrakizumab (100/200 mg SC every 12 weeks; FDA 2018): reSURFACE 1/2 week-12 PASI-75 62–66% versus 6% placebo; the two-year reSURFACE 1 extension reached week-28 PASI-75 78% and PASI-90 58% [[41](#)].

6.2 Icotrokinra (oral macrocyclic peptide; the direct comparator)

Summarized in [Section 3](#): phase 2 week-16 PASI-75 37–79% dose-dependent versus 9% placebo [[6](#)]; phase 2 week-52 PASI-75/90/100 76%/64%/40% and IGA 0 43% [[16](#)]; phase 3 week-16 IGA 0/1

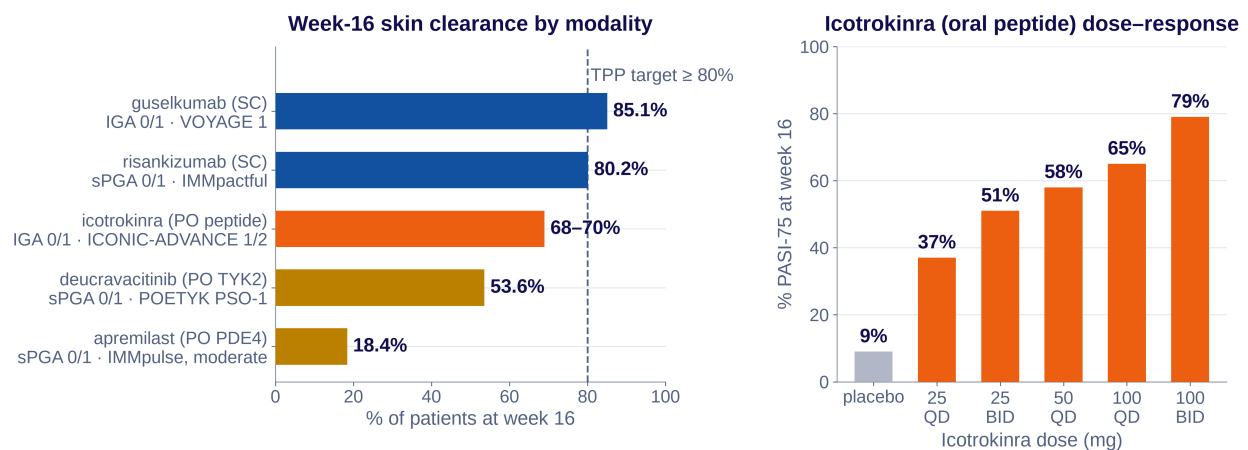
68%/70% versus 11%/9% placebo and PASI-90 55%/57% versus 4%/1% (all $P < 0.0001$, superior to both placebo and deucravacitinib [20]). Phase 3 results are reported; approval status is unverified.

6.3 Oral JAK/PDE4 (the current oral bar)

Deucravacitinib 6 mg QD (oral TYK2 inhibitor): POETYK PSO-1 week-16 PASI-75 58.4% versus 12.7% placebo and 35.1% apremilast; sPGA 0/1 53.6% versus 7.2% and 32.1% [3]; PSO-2 week-16 sPGA 0/1 49.5% [43]. **Apremilast** 30 mg BID (oral PDE4 inhibitor): IMMpulse (phase 4 head-to-head, moderate disease, $n = 352$) week-16 sPGA 0/1 18.4% versus risankizumab 75.4%; PASI-90 5.1% versus 55.9% [42].

Table 3. Week-16 efficacy benchmarks by modality in moderate-to-severe plaque psoriasis (endpoint as listed; dashed line at 80% in Figure 7 is the TPP target).

Agent (class, route)	Benchmark trial	Week-16 efficacy
Guselkumab (anti-IL-23 mAb, SC)	VOYAGE 1 [7]	IGA 0/1 85.1% (placebo 6.9%)
Risankizumab (anti-IL-23 mAb, SC)	IMMpactful [30]	sPGA 0/1 80.2% (deucravacitinib 39.7%)
Icetrokinra (macrocytic peptide, oral)	ICONIC-ADVANCE 1/2 [20]	IGA 0/1 68–70% (placebo 9–11%)
Deucravacitinib (TYK2 inhibitor, oral)	POETYK PSO-1 [3]	sPGA 0/1 53.6% (placebo 7.2%)
Apremilast (PDE4 inhibitor, oral; moderate disease)	IMMpulse [42]	sPGA 0/1 18.4% (risankizumab 75.4%)



Sources: VOYAGE 1 · IMMpactful · ICONIC-ADVANCE 1/2 · POETYK PSO-1 · IMMpulse (moderate disease)

Figure 7. Week-16 skin clearance by modality (left; endpoint per bar): guselkumab 85.1% (subcutaneous mAb, IGA 0/1, VOYAGE 1 [7]); risankizumab 80.2% (subcutaneous mAb, sPGA 0/1, IMMpactful [30]); icetrokinra 68–70% (oral macrocyclic peptide, IGA 0/1, ICONIC-ADVANCE 1/2 [20]); deucravacitinib 53.6% (oral TYK2 inhibitor, sPGA 0/1, POETYK PSO-1 [3]); apremilast 18.4% (oral PDE4 inhibitor, moderate disease, sPGA 0/1, IMMpulse [42]); dashed line at 80% = TPP target. Right: icetrokinra FRONTIER-1 week-16 PASI-75 dose-response (placebo 9%; 25 mg QD 37%, 25 mg BID 51%, 50 mg QD 58%, 100 mg QD 65%, 100 mg BID 79% [6]).

Read together, Figure 7 and Table 3 show that icetrokinra (a peptide) sits 12–15 percentage

points below the mAb bar (80.2–85.1%), and that only oral small molecules—TYK2 and PDE4 inhibitors—currently operate in the 18–54% band. A small molecule at the IL-23R node is judged against the 80% line, not the JAK line.

6.4 Onset of action

A network meta-analysis of 45 trials [35] places time to 50% PASI-90 at approximately 6–8 weeks for IL-17, 9–10 weeks for IL-23, and 11–12 weeks for IL-12/23 blockade, with JAK/PDE4 inhibitors slower. An IL-23 small molecule should not be held to a JAK onset in efficacy readouts; its TPP onset target (week 4, Section 8) is a pharmacologic engagement target, not a clinical-efficacy onset.

6.5 Pharmacokinetics and durability

Half-lives: guselkumab 12–19 days (first-in-human [55]; population pharmacokinetics $\approx$ 14.6 days [46]); risankizumab $\sim$ 28 days with absolute bioavailability 89%, clearance $\sim$ 0.31 L/d, no exposure–safety relationship, no CYP-mediated drug–drug interaction (DDI), and positive exposure–efficacy [33]; tildrakizumab 20.2–28.2 days [53]. Durability extends to week 256 (PASI-90 85.1% [34]). The SC mAbs are dosed every 8–12 weeks; an oral small molecule with a 12–24 h half-life is a different PK problem, not a lower efficacy bar—the week-16 80% line is the same.

7 Safety and class effects

7.1 The IL-23 class is clean

In the 11-trial pooled analysis of guselkumab (4399 patients, 10,787 patient-years [PY]; [44]), the adverse-event rate was 164 per 100 PY, with infections 61.2, serious adverse events (SAE) 5.4, serious infections (SI) 1.0 versus 2.3 for placebo, malignancy 0.6, and major adverse cardiovascular events (MACE) 0.3 per 100 PY; the paper reports no adverse events of Crohn’s disease, ulcerative colitis, or active tuberculosis. Anti-IL-23p19 antibodies are not linked to new-onset IBD in a systematic review of dermatology IL-23 trials [48]—the key differentiation from anti-TNF. Risankizumab has IBD approvals (ulcerative colitis: INSPIRE/COMMAND [28]; Crohn’s disease: SEQUENCE [36]). Anti-drug antibody (ADA) rates are low: guselkumab 3.3–5%, tildrakizumab 6.5–8.2%, risankizumab high-titer <1 –2%; nasopharyngitis/upper respiratory infection is the most common adverse event. Routine latent-tuberculosis-infection (LTBI) screening is not required for IL-17/IL-23 in psoriasis (87.5% consensus, [10]). For context, the IL-17A/F-specific oral candidiasis signal (19.3% bimekizumab versus 3.0% secukinumab, [39]) is the reason the IL-23 class is not expected to carry a candidiasis liability. On suicidality: the only class signal is specific to brodalumab (an IL-17RA-only mAb; 7-year US pharmacovigilance, 5449 patients, $\sim$ 7845 PY; REMS in place; causality not established; no completed suicides, one attempt [26])—no IL-23 signal is reported.

7.2 JAK class baggage (the comparator bar to avoid)

ORAL Surveillance (NEJM 2022 [54]): rheumatoid arthritis patients aged $\geq$ 50 with cardiovascular risk, tofacitinib versus TNF inhibitors (TNFi), median follow-up 4.0 years: MACE 3.4% versus 2.5% (hazard ratio [HR] 1.33, 95% CI 0.91–1.94); cancer 4.2% versus 2.9% (HR 1.48, 95% CI 1.04–2.09); non-inferiority was not demonstrated; herpes zoster (HZ) and opportunistic infections (including TB and non-melanoma skin cancer) were higher. Deucravacitinib’s four-year pooled data (1519 patients, 4392.8 PY [4])—SAE 5.68 $\rightarrow$ 5.01, HZ 0.81 $\rightarrow$ 0.55, malignancy 1.02 $\rightarrow$ 0.89, MACE 0.30 $\rightarrow$ 0.32, venous thromboembolism 0.20 $\rightarrow$ 0.07 per 100 PY—are reassuring within the JAK class but do not exonerate the class. A new oral small molecule must clear this bar by design and by its own data.

7.3 Genetic tolerability

The protective IL23R variant R381Q (rs11209026), associated with reduced risk of both psoriasis and IBD (Duerr 2006; Capon 2007), provides human-genetic evidence that partial loss of IL-23R signaling is tolerated across the immune repertoire, with no IBD phenotype.

7.4 Pharmacodynamic benchmark

The VOYAGE 1 PD substudy [9] shows rapid week-4 reduction of serum IL-17A, IL-17F, and IL-22, sustained to week 48. ECLIPSE [8] shows guselkumab reducing serum IL-17F and IL-22 more than secukinumab at weeks 24/48. The recommended PD battery for a small molecule: serum IL-17A/17F/22 at weeks 1–48, plus ex vivo Th17 cytokine output and pSTAT3/pSTAT4, with the response threshold of >90% suppression in keratinocyte-adjacent lymphocytes [52]; shed sIL-23R as an on-target biomarker [18].

Table 4. Safety benchmarks by class for the IL-23R small-molecule target assessment.

Class	Signal	Benchmark
Anti-IL-23 mAb	SI / MACE / IBD	SI 1.0, MACE 0.3, no IBD AEs (guselkumab pooled, 10,787 PY [44]); no new-onset IBD link [48]
Anti-IL-23 mAb	Candidiasis	Not a class signal (IL-17A/F-specific: 19.3% bimekizumab vs 3.0% secukinumab [39])
Anti-IL-23 mAb	Suicidality	None (signal specific to IL-17RA-only brodalumab [26])
JAK (oral)	MACE / cancer / HZ	Tofacitinib MACE HR 1.33, cancer HR 1.48 vs TNFi (ORAL Surveillance [54]); deucravacitinib 4-yr pooled flat [4]
IL-23 class	LTBI	Routine screening not required (87.5% consensus [10])
IL-23R (genetic)	Tolerability	Protective R381Q variant (psoriasis + IBD; Duerr 2006; Capon 2007)

8 Preclinical target product profile

Table 5 specifies the preclinical bar. Every row is anchored to published benchmark evidence: the mAb efficacy line (80.2–85.1% week-16), the peptide benchmark (icotrokinra 68–70%), the affinity line (IL-23:IL-23R K_D 10–44 nM; icotrokinra 7.1 pM), the selectivity line (Janssen's >3-fold screen floor and icotrokinra's measured 9–18 pM IFN γ IC₅₀), and the safety line (guselkumab pooled SI 1.0 per 100 PY).

The nonclinical package supporting a clinical proof-of-concept:

1. **Species cross-reactivity.** Binding panel across human/mouse/rat/dog IL-23 and IL-23R, plus an hIL-23R transgenic/humanized mouse (human readouts are mandatory; Section 9).
2. **Human ex vivo potency and selectivity.** PBMC/CD4/Th17 functional assays with IL-12, IFN γ , and IL-22 counter-screens.

Table 5. Preclinical target product profile (TPP) for an oral small-molecule IL-23R/p19 antagonist. Benchmark evidence: [7–9, 11, 16, 17, 20, 27, 30, 33–35, 44].

Dimension	Preclinical target	Benchmark evidence
Efficacy (week 16, moderate–severe)	sPGA 0/1 $\geq 80\%$; PASI-90 $\geq 70\%$; PASI-100 $\geq 45\text{--}50\%$	mAb bar 80.2–85.1% [7, 30]; must exceed icotrokinra 68–70% [20]
Complete clearance	PASI-100 $\geq 40\%$ by week 16; stretch $\geq 50\%$ at weeks 48–52	bimekizumab 61.7% [39]; LIM-Mitless 52.3% [34]; icotrokinra week 52 40% [16]
Onset	On-target engagement weeks 1–2; sPGA 0/1 $\geq 30\text{--}40\%$ by week 4	IL-23 mAbs reach 50% PASI-90 at $\sim 9\text{--}10$ weeks [35]
Durability	Sustained at week 52; IGA 0 $\geq 40\%$	Icotrokinra week 52 IGA 0 43% [16]; risankizumab week 256 PASI-90 85.1% [34]
Dosing	Oral once daily, no/low food restriction	Class precedent: oral QD dosing (icotrokinra, deucravacitinib)
Potency	Human Th17 functional $IC_{50} \leq$ low nM; direct binding $K_D \leq \sim 44$ nM	IL-23:IL-23R K_D 10–44 nM [11, 27]; icotrokinra 7.1 pM [17]
Selectivity	No IL-12 impact; $\geq 10\text{--}30\times$ vs IFN γ ; no IL-22 impact	Icotrokinra: no IL-12 effect, IFN γ IC_{50} 9–18 pM [17]; Janssen screen floor $> 3\times$ [49]; IL-35 receptor usage uncertain—test
PD	Serum IL-17A/17F/22 reduced by weeks 1–4 and sustained	VOYAGE 1 substudy [9]; ECLIPSE [8]
Safety	SI $\leq \sim 1.5$ per 100 PY; no IBD, TB, or candidiasis signal; no JAK-class MACE/cancer/HZ	Guselkumab pooled SI 1.0 [44]; ORAL Surveillance [54]
PK	Oral QD, $t_{1/2}$ 12–24 h; positive exposure–efficacy; no exposure–safety; no meaningful CYP DDI; food effect characterized	Risankizumab/guselkumab PK benchmarks [33, 46]

3. **In-vivo proof of concept.** Rat IL-23-induced skin inflammation (thickening plus IL-17A/F/22 gene induction) and rat TNBS colitis as the IBD-safety readout (positive at ≥ 0.3 mg/kg/d for the icotrokinra peptide precedent [17]).
4. **Safety pharmacology and tox.** 28-day (rodent + dog) and 90-day (rat) repeat-dose with recovery; genotoxicity (Ames, micronucleus, chromosome aberration); hERG; drug-induced liver injury (DILI) panel; immunotoxicity with T-cell subset monitoring; developmental/fetal and juvenile studies if the label requires.
5. **Exposure mapping.** NOAEL (no observed adverse effect level) with ≥ 10 -fold margin over projected human exposure, plus exposure–efficacy and exposure–safety mapping.

9 Risks, unknowns, and white space

9.1 Risks (ranked)

1. **Flat-interface chemistry.** The $900 \text{ \AA}^2/51$ -residue pocketless surface is the root cause of slow, hard chemistry; the apo structure has no pocket to exploit directly (Figure 4).
2. **Species translation.** Human–mouse IL-23R cross-reactivity is unverified in the retrievable literature, and the murine and human interfaces differ [15]. Human readouts (PBMC/Th17, hIL-23R transgenic/humanized mice) are mandatory; a rat-only PoC would be insufficient.
3. **Oral exposure unproven for any small molecule.** No published in-vivo small-molecule data exist at this node; the peptide benchmark achieved therapeutic oral exposure at 200 mg QD [20].
4. **Induced-pocket risk.** The 8UUI pocket is peptide-stabilized; no apo or small-molecule-occupied structure of the pocket exists yet. It may close without a peptide.
5. **Glycosylation.** N29 (D1) is unmodeled in every deposited complex; receptor-side leads need a glyco-engineering check before commitment.
6. **Competitive threat.** Icotrokinra phase 3 (68–70% week 16, oral; approval status unverified) and a moving mAb bar (durability to week 256 at PASI-90 85.1% [34]).
7. **IBD ambitions.** Icotrokinra’s IBD phase 2 was terminated for futility (NCT04102111); IBD indications for an IL-23R small molecule carry demonstrated futility risk at the peptide node.

9.2 White space (opportunity)

1. **First-mover structural data.** No small-molecule + IL-23/IL-23R PDB structure exists; the first structure would be a moat.
2. **No covalent option at this interface.** The only IL-23-adjacent covalent small-molecule literature sits at the TYK2 node (Cys736-targeting covalent TYK2 inhibitors [50]); covalent or affinity-catchable chemistry at p19:IL-23R D1 is unclaimed.
3. **p19-only selectivity as a class-first property** (no IL-12 overlap by geometry, Section 2.1).
4. **The induced pocket is a new mechanism** with an unclaimed fragment-to-lead path (180 μM $\rightarrow$ 28 nM headroom).
5. **In-vivo PoC for an oral small molecule at the IL-23R node is unclaimed** (none published in the retrievable literature).

10 Conclusions

The IL-23R–p19 interface is a hard PPI—flat, $\sim 900 \text{ \AA}^2$, with a sub-nanomolar protein ligand to beat—but the 2025 structures prove both halves of the problem: a cryptic pocket is buildable on p19 (8UUI), and the D1:p19 site accepts an oral, pM-affinity binder (9MFG VHH; icotrokinra in vivo). The structural white space is absolute—no small-molecule structure, no covalent option, no in-vivo small-molecule data—which is a first-mover opportunity rather than evidence of inaccessibility. The recommended program is a p19-centric fragment-to-lead campaign (loop-cluster targeting plus induced-pocket exploitation, Section 5) run in parallel with a receptor-side D1 track, gated on the Section 8 TPP: week-16 sPGA 0/1 $\geq 80\%$, oral once-daily dosing with a 12–24 h half-life, onset by week 4, and mAb-class safety ($SI \leq \sim 1.5$ per 100 PY, no IBD signal). The benchmark to beat is not the JAK bar (18–54%) but the mAb bar (80.2–85.1%); the peptide competitor (68–70%) proves the route is chemically reachable.

11 Data availability and methods

This report was compiled exclusively from the five project source records:

- handoff_notes_IL23R_deck.md
- brief_IL23_IL23R_structural_biology.md
- brief_IL23R_small_molecule.md
- TPP_brief_IL23R_oral_psoriasis.md
- deck/build_deck.py

Every quantitative claim is traceable to one of these files. Public retrieval during the project used NCBI E-utilities (esearch/esummary/efetch) for reference metadata, Europe PMC for full text, and the RCSB entry-data API; structures were verified against RCSB coordinates. ClinicalTrials.gov, openFDA, Wikipedia, Google Scholar, ChemRxiv, and the patent databases (Google Patents, EPO Espacenet, USPTO, PatentsView, FreepatentsOnline) returned HTTP 403 at query time; all white-space claims are therefore qualified “in the retrievable literature” and do not exclude unpublished or paywalled patent claims. Known-false claims circulating in earlier drafts were excluded: the 6B0G/6B1K–6B1R entries as Schreiber-lab small-molecule IL-23 inhibitors (false); a 2004 *Nature* IL-23 structure (the first IL-23 structures are 2008, 3DUH/3D87 [5, 29]); “IL-23R has four CRDs” (it has one Ig D1 and two Fn3 domains); and “p19 has CRDs” (p19 is a four-helix-bundle class II cytokine with no domains). Items flagged UNCERTAIN in the sources (exact CDR3 boundary, N29 glycan role, IL-35 receptor usage) are stated as such and not used as design facts. The two genome-wide association references (Duerr 2006; Capon 2007) are named in the source record without PMIDs; they appear in the text without numbered citations and are not in the reference list. Figures are rendered from PDB coordinates with the project’s frozen chain colors; buried-surface-area values are as reported in the primary papers, not recomputed.

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